Equilibrium Propagation with Natural Frequencies in Coupled Oscillator Networks

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Abstract

We prove that in a coupled Kuramoto oscillator network at stable equilibrium, the physical phase displacement under weak output nudging is the gradient of the loss with respect to natural frequencies, with equality as the nudging strength $\beta \to 0$. Prior oscillator equilibrium propagation work explicitly set aside natural frequency as a learnable parameter [Wang et al., 2024]; we show that on sparse layered architectures, frequency learning outperforms coupling-weight learning among converged seeds (96.0% vs. 83.3% at matched parameter counts, $p < 10^{-12}$). The ∼50% convergence failure rate under random initialization is a loss-landscape property, not a gradient error; topology-aware *spectral seeding* eliminates it in all settings tested (46/100 → 100/100 seeds on the primary task; 50/50 on a second task, K-only training, and a larger architecture).

1 Introduction

The digital infrastructure for training neural networks is built on a single assumption: that computing parameter gradients requires a computation distinct from, and more expensive than, inference. The backward pass of backpropagation stores every intermediate activation, runs the transpose Jacobian layer by layer, and coordinates globally via the chain rule. For a transformer with L layers and hidden dimension d, the backward pass costs approximately $4Nd$ floating-point operations per token—twice the forward pass—and requires storing $O(L \cdot d)$ activations in high-bandwidth memory [Korthikanti et al., 2023]. At the scale of GPT-3, this means 1,287 MWh of training energy and ∼118 GiB of activation storage per training step [Patterson et al., 2021].

The machine learning community is aware of this cost. The response has been a proliferation of digital workarounds: quantization to 1.58-bit ternary weights eliminates matrix multiplication [Ma et al., 2024]; state-space models replace quadratic attention with linear dynamics [Gu and Dao, 2023]; mixture-of-experts activates only 2% of parameters per token [Shazeer et al., 2017]; KV-cache compression reduces memory from $O(N)$ to constant size [Liu et al., 2024]; and energy-based transformers replace single-pass softmax with iterative gradient descent on an energy surface, achieving better out-of-distribution generalization in 3 forward passes [Gladstone et al., 2025]. Each of these techniques addresses a specific bottleneck of the digital training pipeline.

We show that coupled oscillator networks sidestep one of these bottlenecks entirely: the gradient of a loss function with respect to oscillator parameters is physically encoded in the equilibrium phase response, requiring no backward pass, no stored activations, and no chain rule.

Specifically, we prove a *phase-gradient identity* for Kuramoto oscillator networks: weakly nudging the output toward a target causes every oscillator's phase to shift by exactly the loss gradient with respect to its natural frequency (Theorem 1, Figure 1). The proof reveals that the Jacobian of the Kuramoto equilibrium is a coupling-weighted graph Laplacian, connecting

gradient propagation to diffusion on the coupling graph. This identity is an instance of equilibrium propagation [Scellier and Bengio, 2017] applied to Kuramoto dynamics with natural frequency as the learnable parameter—a parameter that all prior oscillator EP work explicitly set aside.

This paper makes three contributions:

1. **The phase-gradient identity** (Section 4). We derive the exact gradient formula for natural frequencies via the implicit function theorem, revealing the Laplacian structure of the equilibrium Jacobian. The identity is verified to machine precision (cosine similarity = 1.000000) across networks of 6 to 200 oscillators using three methodologically independent cross-checks (two-phase, analytical, and finite-difference; all share a common equilibrium solver).

2. **Conditional advantage of natural frequency on sparse architectures** (Section 6.3). Across 100 seeds, all-seed accuracy is indistinguishable between ω-only and K-only ($p = 0.98$). Conditioned on training convergence, ω-only achieves 96.0% vs. 83.3% at matched parameter counts (7 vs. 7; $p < 10^{-12}$, Welch's t-test). Tripling K's parameter budget does not close the gap. This is a sparse-architecture result; coupling-only learning achieves state-of-the-art accuracy on dense networks [Wang et al., 2024, Rageau and Grollier, 2025].

3. **Convergence diagnosis and spectral seeding** (Section 6.4). The ∼50% convergence failure rate under random initialization is consistent with loss-landscape multistability, not gradient error: the same seeds fail across five distinct training configurations (Cochran's $Q = 0$). In the sparse layered settings tested, we eliminate this failure via *spectral seeding*: initializing ω from the eigenvectors of the coupling graph Laplacian, weighted by output separation and inverse eigenvalue. This raises the fraction of seeds exceeding 60% final test accuracy from 46/100 to 100/100 on the primary task, and achieves 50/50 across a second vowel task, K-only training, and a larger architecture.

This work connects three lines that have not previously been unified. Hoppensteadt and Izhikevich [1999] showed that Kuramoto oscillator networks have Hopfield-like associative memory properties and proposed oscillator hardware, but used only Hebbian learning. Scellier and Bengio [2017] proved that equilibrium propagation computes exact gradients in energy-based systems, but the framework does not identify which parameters to learn or reveal the gradient structure for specific dynamics. Wang et al. [2024] applied EP to phase oscillators but explicitly set aside natural frequency as a learnable parameter. We complete the arc: the gradient that Hoppensteadt's oscillators could not compute, that Scellier's framework guarantees exists, and that Wang's implementation set aside, was in the phase displacement all along.

2 Related Work

Equilibrium propagation. Scellier and Bengio [2017] proved that for energy-based models at equilibrium, the gradient of a cost function with respect to model parameters can be estimated from the difference between a free-running and a weakly nudged equilibrium. Scellier [2018] later generalized EP to vector field dynamics without an energy function. Ernoult et al. [2019] proved that EP gradients match backpropagation-through-time gradients exactly in RNNs with static input. Zucchet and Sacramento [2022] unified EP with implicit differentiation in a bilevel optimization framework, showing that EP is a physical implementation of the general IFT-based gradient computation. Our derivation is an instance of this framework applied to Kuramoto dynamics; the value of the specific instantiation is that it reveals the Jacobian-as-Laplacian structure, identifies ω as a learnable parameter that prior work explicitly set aside, and enables machine-precision verification of gradient exactness.

EP in oscillator networks. Wang, Wanjura, and Marquardt [2024] applied EP to XY-coupled phase oscillators, optimizing coupling weights W_{ij} and bias phases ψ_i on XOR and MNIST tasks (94.1% on 8×8 MNIST with a layered architecture). They explicitly set aside natural frequencies, noting that frequency dispersion "may lead to the absence of stable fixed points." Rageau and Grollier [2025] extended oscillator EP to handle frequency dispersion, achieving 97.8% on full MNIST in the synchronized case and 96–97% under dispersion, with coupling-only learning. Gower et al. [2025a, 2025b] mapped EP onto oscillator Ising machines in standard CMOS. Zoppo et al. [2022] applied EP to memristor-based Van der Pol oscillator networks for associative memory, training coupling weights in a circuit-realistic setting. None of these works derive the gradient formula for natural frequency or verify gradient exactness.

Onsager reciprocity and EP. Wanjura and Marquardt [2025], working in the context of quantum systems, showed that Onsager reciprocity provides the foundation for EP: in any system with reciprocal interactions, a single nudge experiment at the output yields gradients for all parameters simultaneously. Our identity is the Kuramoto instantiation of this principle, with the additional finding that natural frequency—a parameter explicitly set aside by prior oscillator EP work—is the more effective one on sparse architectures.

Oscillatory neurocomputers. Hoppensteadt and Izhikevich [1999] showed that Kuramoto oscillator networks have Hopfield-like associative memory properties and proposed hardware implementations using voltage-controlled oscillators, Josephson junctions, and MEMS. Their networks compute via phase but learn only through Hebbian pattern storage, not gradient-based training. Our work completes their vision: the gradient identity enables gradient-based training of the oscillator parameters they proposed for hardware 27 years ago.

Contrastive Hebbian learning and resistive networks. Xie and Seung [2003] proved that contrastive Hebbian learning computes the same gradients as backpropagation in layered networks. Kendall et al. [2020] applied EP to train nonlinear resistor networks, where voltage displacement encodes conductance gradients—the DC-circuit analog of our phase-gradient identity. Dillavou et al. [2024] demonstrated contrastive learning in physical resistive networks without a processor.

Physical learning machines. Momeni et al. [2025] provide a comprehensive framework for training physical neural networks, projecting energy advantages of up to four orders of magnitude. All such projections lack empirical calibration at matched process nodes.

Deep equilibrium models. Bai, Kolter, and Koltun [2019] introduced deep equilibrium models (DEQs), which find fixed points of infinite-depth networks and differentiate through them via the implicit function theorem. Our derivation uses the same mathematical machinery but applies it to physical oscillator dynamics rather than abstract neural network layers.

Oscillator computing. Todri-Sanial et al. [2024] survey oscillator-based computing across pattern recognition, combinatorial optimization, and machine learning. Their taxonomy provides the hardware context for our contribution. Miyato et al. [2025] use a vector-valued generalization of Kuramoto dynamics as a drop-in activation function within standard neural networks, trained end-to-end by backpropagation. Their work demonstrates the computational utility of oscillator dynamics at scale but does not derive frequency gradients via EP or address which oscillator parameters to learn.

Digital efficiency techniques. The machine learning community has developed extensive workarounds for the cost of digital training: ternary weights eliminate matrix multiplication [Ma et al., 2024]; state-space models replace quadratic attention with linear dynamics [Gu and Dao, 2023]; energy-based transformers perform iterative energy descent rather than single-pass prediction [Gladstone et al., 2025]. Each addresses a specific digital bottleneck. Coupled oscillator networks avoid some of these bottlenecks by construction (e.g., no stored activations, no sequential backward pass), though whether this translates to practical advantage is an open question that requires hardware validation beyond the scope of this paper.

3 Background

3.1 Kuramoto Oscillator Networks

Consider a network of N coupled phase oscillators with dynamics

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \sum_{j=1}^{N} K_{ij} \sin(\theta_j - \theta_i), \quad i = 1, \ldots, N, \tag{1}$$

where $\theta_i \in [0, 2\pi)$ is the phase of oscillator i, $\omega_i \in \mathbb{R}$ is its natural frequency, and $K_{ij} \geq 0$ is the coupling strength from oscillator j to oscillator i [Kuramoto, 1984, Strogatz, 2000].

When the coupling is sufficiently strong relative to the frequency spread, the network synchronizes: all oscillators lock to a common frequency, and there exists a stable equilibrium $\boldsymbol{\theta}^*$ in a rotating frame satisfying

$$F_i(\boldsymbol{\theta}^*, \boldsymbol{\omega}) \;\coloneqq\; \omega_i^c + \sum_{j=1}^{N} K_{ij} \sin(\theta_j^* - \theta_i^*) = 0, \quad i = 1, \ldots, N, \tag{2}$$

where $\omega_i^c = \omega_i - \bar{\omega}$ are the mean-centered frequencies and $\bar{\omega} = \frac{1}{N}\sum_i \omega_i$.

Remark 1 (Rotational symmetry)**.** *The equilibrium has a continuous symmetry: if $\boldsymbol{\theta}^*$ is a solution, so is $\boldsymbol{\theta}^* + c\mathbf{1}$ for any constant c. We break this by pinning $\theta_1^* = 0$ and solving the reduced $(N-1)$-dimensional system.*

3.2 The Jacobian as a Graph Laplacian

The Jacobian of F with respect to $\boldsymbol{\theta}$ at equilibrium is

$$J_{ij} = \left.\frac{\partial F_i}{\partial \theta_j}\right|_{\boldsymbol{\theta}^*} = \begin{cases} K_{ij}\cos(\theta_j^* - \theta_i^*), & i \neq j, \\ -\sum_{\ell \neq i} K_{i\ell}\cos(\theta_\ell^* - \theta_i^*), & i = j. \end{cases} \tag{3}$$

This is a *coupling-weighted graph Laplacian*: the off-diagonal entry J_{ij} is the effective coupling strength at equilibrium, and each row sums to zero. For a connected graph with symmetric coupling, the reduced Laplacian $\tilde{J}$ (with the pinned node removed) is positive definite by Fiedler's theorem [Chung, 1997], guaranteeing the nonsingularity required by the implicit function theorem.

Proposition 1 (Symmetry)**.** *If $K_{ij} = K_{ji}$ for all i, j, then J is symmetric, and so is $\tilde{J}$.*

Proof. $J_{ij} = K_{ij}\cos(\theta_j^* - \theta_i^*) = K_{ji}\cos(\theta_i^* - \theta_j^*) = J_{ji}$, since cosine is even and $K_{ij} = K_{ji}$. □

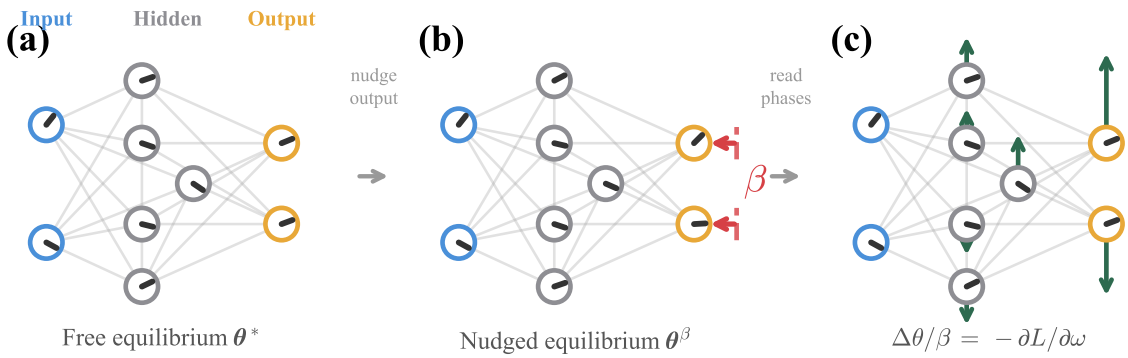

Figure 1: The two-phase gradient readout protocol. **(a)** Solve for the free equilibrium $\boldsymbol{\theta}^*$. **(b)** Add a weak nudging force $-\beta(\theta_i - \theta_i^{\text{tgt}})$ at output oscillators and re-solve for $\boldsymbol{\theta}^\beta$. **(c)** The per-node phase displacement $(\theta_k^\beta - \theta_k^*)/\beta$ approximates the negative gradient $-\partial L/\partial\omega_k^c$ at finite β, with equality in the limit $\beta \to 0$ (Theorem 1). Blue: input nodes (frequencies set by data). Gray: hidden nodes. Amber: output nodes. Dashed line in (b): target phase.

3.3 Equilibrium Propagation

Equilibrium propagation [Scellier and Bengio, 2017] computes parameter gradients in energy-based systems at fixed points. For a system with energy $E(\boldsymbol{\theta}; \lambda)$ and output loss $L(\boldsymbol{\theta})$, weakly nudging the output produces a perturbed equilibrium whose displacement encodes the parameter gradient:

$$\frac{\partial L}{\partial \lambda} = \lim_{\beta \to 0} \frac{1}{\beta} \left[\frac{\partial E}{\partial \lambda}\bigg|_{\boldsymbol{\theta}^\beta} - \frac{\partial E}{\partial \lambda}\bigg|_{\boldsymbol{\theta}^*} \right]. \tag{4}$$

The Kuramoto dynamics (1) can be written as gradient descent on the energy $E(\boldsymbol{\theta}) = -\sum_i \omega_i \theta_i - \sum_{i<j} K_{ij} \cos(\theta_j - \theta_i)$, so EP applies. However, EP does not specify *which* parameters are effective targets for optimization, does not reveal the Jacobian structure, and does not guarantee that gradients with respect to natural frequencies will be well-behaved. The next section addresses these questions.

4 Phase-Gradient Identity

4.1 Setup

We partition the oscillators into input (frequencies set by data), hidden, and output nodes. The coupling matrix K is symmetric with non-negative entries.

Definition 1 (Output loss)**.** $L(\boldsymbol{\theta}^*) = \frac{1}{2} \sum_{i \in \mathcal{O}} (\theta_i^* - \theta_i^{\text{tgt}})^2$, *where $\mathcal{O}$ is the set of output oscillators.*

Definition 2 (Nudged equilibrium)**.** *For nudging strength $\beta > 0$, $\boldsymbol{\theta}^\beta$ satisfies*

$$\omega_i^c + \sum_j K_{ij} \sin(\theta_j^\beta - \theta_i^\beta) - \beta\,(\theta_i^\beta - \theta_i^{\text{tgt}})\,\mathbf{1}_{i \in \mathcal{O}} = 0. \tag{5}$$

Assumption 1. *The free equilibrium $\boldsymbol{\theta}^*$ exists, is stable, and the reduced Jacobian $\tilde{J}$ is nonsingular. By Fiedler's theorem, this holds whenever the coupling graph is connected and all effective coupling strengths $K_{ij}\cos(\Delta\theta^*_{ij})$ are positive.*

4.2 Main Result

Theorem 1 (Phase-gradient identity)**.** *Under Assumption 1, with symmetric coupling, for every unpinned oscillator k:*

$$\lim_{\beta\to 0}\frac{\theta_k^\beta-\theta_k^*}{\beta}=-\frac{\partial L}{\partial\omega_k^c}, \tag{6}$$

where $\omega_k^c=\omega_k-\bar{\omega}$ is the mean-centered frequency.

Proof. The energy (4) depends on ω_k only through the term $-\omega_k\theta_k$, so $\partial E/\partial\omega_k=-\theta_k$. Applying the general EP formula (4) with $\lambda=\omega_k$:

$$\frac{\partial L}{\partial\omega_k}=\lim_{\beta\to 0}\frac{1}{\beta}\left[-\theta_k^\beta-(-\theta_k^*)\right]=-\lim_{\beta\to 0}\frac{\theta_k^\beta-\theta_k^*}{\beta}. \qquad \square$$

Remark 2 (Structural interpretation via the IFT)**.** *The identity also follows from the implicit function theorem, which reveals the mechanism of gradient propagation. The equilibrium condition $F(\boldsymbol{\theta}^*,\boldsymbol{\omega})=0$ gives $\partial\boldsymbol{\theta}^*/\partial\boldsymbol{\omega}^c=-\tilde{J}^{-1}$,*

$$\frac{\partial\boldsymbol{\theta}^*}{\partial\boldsymbol{\omega}^c}=-\tilde{J}^{-1}, \tag{7}$$

since $\partial F/\partial\boldsymbol{\omega}^c=I$. The loss gradient is then $\partial L/\partial\boldsymbol{\omega}^c=-\tilde{J}^{-\mathsf{T}}\boldsymbol{e}$, where $e_i=(\theta_i^*-\theta_i^{\mathrm{tgt}})\mathbf{1}_{i\in\mathcal{O}}$, and the nudge response is $\mathrm{d}\boldsymbol{\theta}^\beta/\mathrm{d}\beta|_{\beta=0}=\tilde{J}^{-1}\boldsymbol{e}$. By Proposition 1, $\tilde{J}^{-\mathsf{T}}=\tilde{J}^{-1}$, so the two expressions coincide. This derivation shows that gradient propagation is mediated by $\tilde{J}^{-1}$, the Green's function of the coupling-weighted graph Laplacian: the gradient diffuses on the coupling graph.

Remark 3 (Coupling weight gradient)**.** *The same two equilibria also yield coupling gradients: $\partial L/\partial K_{ij}=\lim_{\beta\to 0}\beta^{-1}[\cos(\theta_j^*-\theta_i^*)-\cos(\theta_j^\beta-\theta_i^\beta)]$, from the general EP formula (4).*

Remark 4 (Mean-centering)**.** *The identity holds for mean-centered frequencies ω^c. Perturbing a raw ω_k shifts the mean, introducing a rank-1 correction. In practice, we perturb centered frequencies directly.*

Remark 5 (Asymmetry)**.** *The identity requires $\tilde{J}^{-\mathsf{T}}=\tilde{J}^{-1}$, which holds iff K is symmetric. In CMOS, capacitive coupling is inherently symmetric. For asymmetric coupling, the discrepancy degrades gracefully: cosine similarity between the two-phase gradient and the true gradient exceeds 0.995 at 20% asymmetry (Table 2).*

4.3 Finite-β Error

At finite β, the error is $O(\beta)$. At the training value $\beta=0.1$, the gradient direction is correct (cosine >0.999) with a mean scale error of 1.3% across 20 random networks.

5 Numerical Verification

We compute $\partial L/\partial\boldsymbol{\omega}^c$ by four methods: (1) **two-phase**: $-(\boldsymbol{\theta}^\beta-\boldsymbol{\theta}^*)/\beta$ at $\beta=10^{-4}$; (2) **analytical**: $-\tilde{J}^{-1}\boldsymbol{e}$; (3) **finite-difference**: perturb each ω_k^c by $\pm 10^{-5}$, re-solve, compute centered difference; (4) **autograd**: reimplemented the equilibrium solver as a Newton iteration in PyTorch, differentiated through it via the implicit function theorem using a custom

Table 1: Phase-gradient identity verification. Cosine similarity between gradient vectors computed by the scipy-based two-phase (TP) and finite-difference (FD) methods, and by the computationally independent PyTorch autograd (AG) method.

N	Freq. params	Cosine (TP vs FD)	Cosine (AG vs TP)	Eq. residual
6	5	1.000000	1.000000	8.4×10^{-14}
10	9	1.000000	1.000000	2.2×10^{-16}
15	14	1.000000	1.000000	3.4×10^{-14}
20	19	1.000000	1.000000	2.9×10^{-15}
30	29	1.000000	1.000000	1.0×10^{-15}
50	49	1.000000	1.000000	8.3×10^{-17}
100	99	1.000000	1.000000	1.1×10^{-16}
200	199	1.000000	1.000000	1.1×10^{-16}

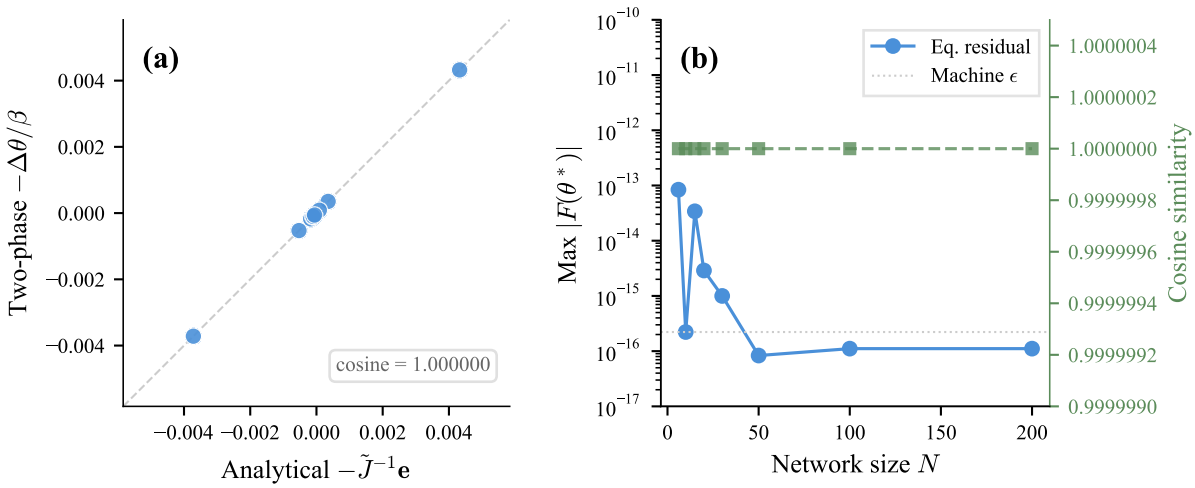

Figure 2: Gradient identity verification. **(a)** Per-node scatter of the two-phase gradient vs. the analytical gradient at $N = 15$, $\beta = 10^{-3}$: all points fall on $y = x$ (cosine similarity = 1.000000). **(b)** Equilibrium residual across network sizes from $N = 6$ to $N = 200$; all residuals are at or below machine ϵ. The cosine similarity is 1.000000 at every scale.

`torch.autograd.Function`, and obtained $\partial L/\partial \boldsymbol{\omega}^c$ by backpropagation. Methods (1)–(3) share the equilibrium solver (`scipy.optimize.fsolve`); method (4) is computationally independent, using only PyTorch tensors and `torch.linalg.solve` for both the forward solve and the backward pass.

Table 1 shows the results across random Erdős–Rényi networks (edge probability 0.6, $K_{\text{mean}} = 5.0$, $\omega \sim \mathcal{N}(0, 0.3^2)$). At every size from $N = 6$ to $N = 200$, the cosine similarity between two-phase and finite-difference gradients is 1.000000 (Figure 2a). All four pairwise cosines agree. Crucially, the autograd column (AG vs TP) provides a computationally independent check: the PyTorch Newton solver and implicit-differentiation backward pass share no code with the scipy-based methods, yet produce identical gradients at every scale. Multi-seed tests (10 seeds at $N = 10$) confirm robustness, with all cosines exceeding 0.999999.

Table 2: Asymmetry robustness ($N = 15$, 10 seeds per level).

Coupling asymmetry	Cosine (TP vs FD)
0%	1.000000 ± 0.000000
5%	0.999766 ± 0.000230
10%	0.999026 ± 0.001001
20%	0.995767 ± 0.004742
50%	0.965372 ± 0.046598

Table 3: Parameter ablation (100 seeds, 150 epochs, lr = 0.001). Convergence defined by final-epoch training accuracy > 60%; reported accuracy is final-epoch test accuracy. Features normalized using training-set statistics only.

Task	Params	Conv. rate	Conv. test acc.	All-seed test
	ω only	47% (47/100)	94.3%	48.9%
/a/ vs /i/	K only	48% (48/100)	82.3%	48.7%
	Both	48% (48/100)	92.2%	49.4%

6 Learning Experiments

6.1 Task and Architecture

We train a Kuramoto network on binary vowel classification using the Hillenbrand et al. [1995] dataset (formant frequencies F_1 and F_2; 275 samples for /a/ vs /i/, drawn from 689 total across five vowels). The network has $N = 9$ oscillators: 2 input, 5 hidden, 2 output, yielding 7 learnable natural frequencies and 24 coupling weights. Classification uses $\hat{c} = \arg\max_c \cos(\theta^*_{\text{out}_c})$: the output oscillator whose phase is closest to zero determines the class.

6.2 Gradient Rule Validation

We compare ω-only two-phase training with ω-only finite-difference training across 10 seeds (same architecture, data, and learning rate). The mean accuracy gap is $0.7\% \pm 0.9\%$, with 6 of 10 seeds producing exactly 0.0% gap. The two-phase rule produces a learning signal functionally equivalent to numerical differentiation.

6.3 Parameter Ablation

Wang et al. [2024] state that frequency terms "are not helpful." We first compare at unmatched parameter counts, then control for this confound in a parameter-matched experiment (Table 4). Table 3 and Figure 3 test this directly. On both tasks, ω-only matches joint training and outperforms K-only in the 20-seed pilot. A 100-seed replication on /a/ vs /i/ confirms the gap among training-converged seeds (final-epoch training accuracy $>60\%$): ω-only 94.3% (47/100 converged) vs. K-only 82.3% (48/100 converged), $p = 1.5 \times 10^{-10}$ by Welch's t-test. The convergence rate difference is not significant ($p = 1.0$, Fisher's exact test). Across all 100 seeds (including non-converged), mean test accuracy is 48.9% vs. 48.7% ($p = 0.98$); the bimodal distribution ($\sim$50% of seeds at chance) dominates the all-seed comparison. Features are normalized using training-set statistics only; we report final-epoch test accuracy throughout.

A symmetric learning-rate sweep across five values (10^{-4} to 10^{-2}) shows ω dominates K at every learning rate: best ω is 96.6% at lr = 0.01 (15/20 converge); best K is 82.2% at lr = 0.001 (9/20 converge).

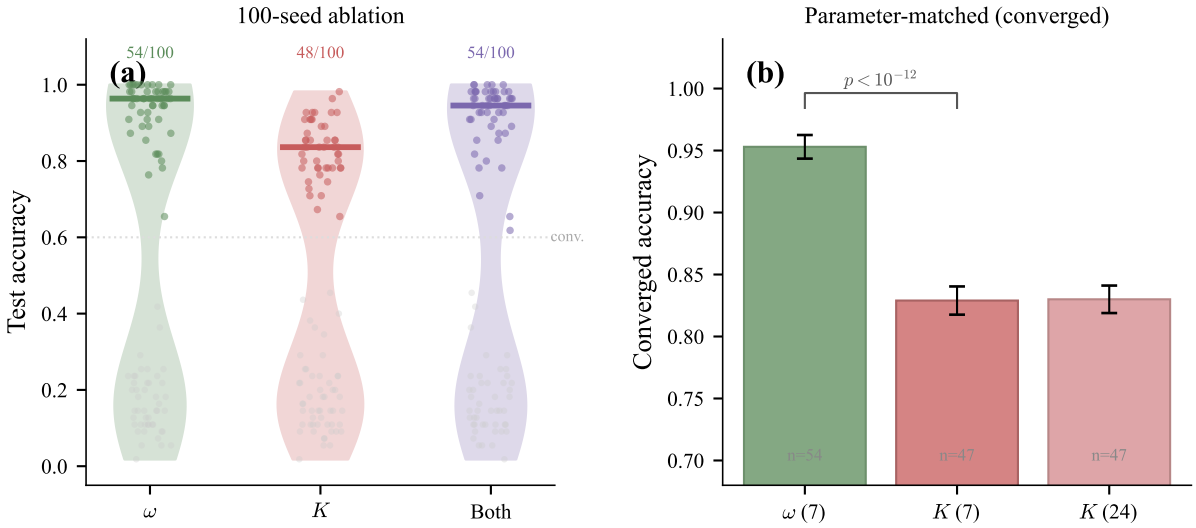

Figure 3: Parameter ablation on /a/ vs /i/ (100 seeds, sparse layered 2+5+2 topology). **(a)** Violin plots of all 100 seeds showing the bimodal distribution: 47/100 ω-only seeds and 48/100 K-only seeds training-converge (final-epoch training accuracy above 60%; convergence rate difference not significant, Fisher $p = 1.0$). Purple: joint ($\omega + K$) training. **(b)** Training-converged seeds only: ω-only achieves 94.3% vs. K-only 82.3% ($p = 1.5 \times 10^{-10}$, Welch's t-test).

Table 4: Parameter-matched ablation (100 seeds, 200 epochs, lr = 0.001). Convergence defined by final-epoch training accuracy > 60%. At equal parameter counts (7 vs. 7), ω-only outperforms K-matched by 12.7 percentage points. Tripling K's parameter budget does not close the gap.

Condition	Params	Conv. rate	Conv. test acc.
ω-only	7	47% (47/100)	96.0%
K-matched	7	46% (46/100)	83.3%
K-full	24	47% (47/100)	83.0%

On a binary FM frequency-discrimination task (autocorrelation features, 22 oscillators), the gap widens: ω-only achieves 73.4% with 10/20 seeds converging; K-only achieves 52.2% with 1/20 converging—at chance. Adding K to ω provides no benefit on either task.

Parameter-matched control. The original ablation compares 7 frequency parameters against 24 coupling parameters. To control for the parameter-count confound, we repeat the comparison at matched parameter counts: for K-matched, we randomly select 7 of the 24 edges to learn and freeze the rest (100 seeds, 200 epochs). Table 4 shows the result: ω-only achieves 96.0% converged test accuracy vs. K-matched 83.3% at equal parameter count ($p = 1.8 \times 10^{-12}$, Welch's t-test; convergence defined by final-epoch training accuracy exceeding 60%). Head-to-head on all 100 seeds, ω wins on 46 vs. 53 for K ($p = 0.90$, Wilcoxon); the all-seed comparison is not significant because ∼50% of seeds remain at chance in both conditions. Giving K its full 24 parameters does not help: K-full achieves 83.0%, indistinguishable from K-matched at 83.3%.

An architecture sweep across five network sizes (Table 5) confirms that ω achieves higher converged accuracy at every scale, with the advantage largest in the sparse regime (13.1 points at N=7) and narrowing as coupling parameter count grows (1.7 points at N=19).

Table 5: Architecture sweep (50 seeds per size, 200 epochs). ω-only outperforms K-only at every network size despite having 2–4× fewer parameters.

N	ω params	K params	ω conv. acc.	K conv. acc.	Gap
7	5	14	96.2%	83.1%	+13.1
9	7	24	94.2%	81.8%	+12.4
11	9	34	91.0%	82.9%	+8.1
14	12	49	89.3%	82.7%	+6.6
19	17	74	88.0%	86.3%	+1.7

Table 6: Convergence diagnosis (40 seeds, 150 epochs). The gradient identity holds for all seeds, including those that fail to learn.

	Converged (18/40)	Failed (22/40)
Init cond($\tilde{J}$)	15.2 ± 1.9	15.4 ± 1.8
Final cond($\tilde{J}$)	14.5 ± 1.6	14.7 ± 1.5
Skip rate	0.0	0.0

Scope. The conditional $\omega > K$ finding is specific to sparse layered architectures and to training-converged seeds; the all-seed comparison is not significant. Wang et al. achieve 94.1% on MNIST with K-only on dense networks, and Rageau and Grollier achieve 97.8%. The advantage of ω emerges when coupling has too few parameters to encode the function and frequency provides a more efficient parameterization. The point is not that ω universally dominates K, but that prior work set aside ω without testing it in the sparse regime where, conditioned on convergence, it is the stronger parameter.

6.4 Convergence Failure Analysis

Approximately 50% of random initializations fail to converge on the vowel task. We tracked the Jacobian condition number, equilibrium residual, and sample skip rate across 40 seeds for 150 epochs (Table 6, Figure 4).

The condition number and skip rate are identical for converged and failed seeds. In a separate 20-seed experiment, five training configurations (baseline, stronger coupling at K=4, coupling floor at K≥1.5, frequency recentering, and ω-only with fixed K=3) produce the exact same convergence pattern: the same 11 seeds converge and the same 9 fail (Cochran's $Q = 0$, degenerate). Because the experiment uses a single seed for both network initialization and data split, we cannot isolate which factor determines the boundary; we can only say that the outcome is fully determined by the seed index and invariant to the training algorithm. This is consistent with multistability of the Kuramoto energy landscape [Wang et al., 2024].

This diagnosis points toward initialization rather than gradient estimation as the missing ingredient: the problem is "find the right basin," not "fix the gradient."

Spectral seeding. In the 2+5+2 architecture, both output oscillators connect to all five hidden oscillators—they are topologically symmetric. The *only* mechanism to break this symmetry is through natural frequencies. Random initialization has roughly a coin-flip chance of placing ω in a basin where the equilibrium separates the output phases in a class-discriminative way.

We exploit the spectral structure of the coupling graph. Let v_i, λ_i be the eigenvectors and eigenvalues of the reduced graph Laplacian $\tilde{L} = \tilde{D} - \tilde{K}$ (the Jacobian at $\boldsymbol{\theta} = 0$, since $\cos(0) = 1$).

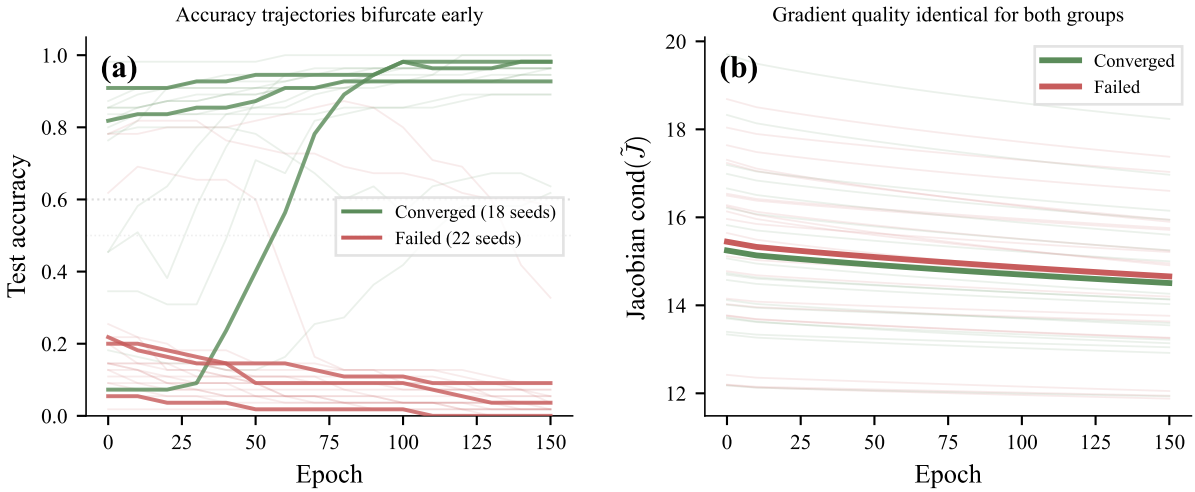

Figure 4: Convergence failure is a landscape problem, not a gradient problem. A converging seed (green) and a failing seed (amber) show different accuracy trajectories but indistinguishable diagnostics: the Jacobian condition number stays ≈ 15 and the skip rate is zero for both seeds.

Table 7: Spectral seeding vs. random initialization (100 seeds, ω-only, 200 epochs). Success defined as final test accuracy exceeding 60%.

Initialization	Success rate	Success accuracy	Overall
Random	46/100 (46%)	$96.8\% \pm 5.1\%$	48.5%
Spectral seed	100/100 (100%)	$97.6\% \pm 2.8\%$	97.6%

Define the signed output separation of mode i as $s_i = [v_i]_{\text{out}_1} - [v_i]_{\text{out}_2}$. We initialize

$$\omega_k = \alpha \sum_i \frac{s_i}{\lambda_i} [v_i]_k, \tag{8}$$

where α normalizes the maximum frequency to 0.3. Input frequencies are set to zero (overwritten per sample). This formula weights each eigenmode by how well it separates the output nodes (s_i) and how strongly the inverse Laplacian amplifies it ($1/\lambda_i$), distributing frequency across *all* nodes in a pattern aligned with the coupling topology.

Table 7 and Figure 5 show the result: spectral seeding raises the success rate from 46/100 to 100/100 seeds, rescuing 54 seeds that fail under random initialization and losing none. The accuracy among successful seeds is unchanged (97.6% vs. 96.8%). The cost is one eigendecomposition of the $(N-1) \times (N-1)$ reduced Laplacian—negligible relative to training.

Alternative initialization strategies that concentrate frequency on output nodes alone (setting $\omega_{\text{out}} = \pm 0.3$ with hidden frequencies near zero) achieve 0% success, confirming that the frequency pattern must span the entire network, not just the readout layer. Multi-start screening (10 random initializations, selecting by initial output separation) provides no improvement over single random initialization, confirming that static output separation does not predict success.

Generalization. Spectral seeding is architecture-aware: it uses the topology of K and the identity of the output nodes. To test breadth, we repeat the experiment under three variations (50 seeds each): a different task (/o/ vs /u/), K-only training (spectral seeding sets the initial ω, which is then frozen while K trains—this shows that a good frequency initialization enables K-only refinement, not that K-only learning is intrinsically fixed), and a larger architecture (2+8+2, N=12). In all three cases spectral seeding achieves 50/50 vs. 19–22/50 for random

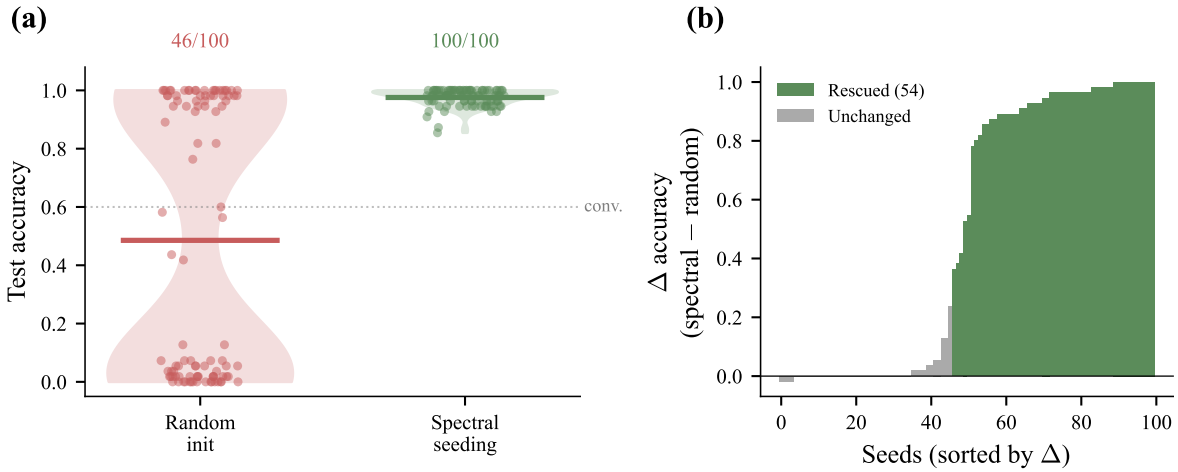

Figure 5: Spectral seeding eliminates the random-initialization basin failure (100 seeds, ω-only, 200 epochs). **(a)** Random initialization produces a bimodal distribution: 46/100 seeds exceed 60% test accuracy. Spectral seeding places all 100 seeds above 90%. **(b)** Per-seed accuracy change: 54 seeds rescued (green), 0 lost.

Table 8: Spectral seeding generalization (50 seeds per setting, 200 epochs).

Setting	Init	Success	Acc.
/o/ vs /u/, ω-only, 2+5+2	Random	19/50	76.5%
	Spectral	50/50	77.9%
/a/ vs /i/, K-only, 2+5+2	Random	22/50	81.4%
	Spectral	50/50	84.6%
/a/ vs /i/, ω-only, 2+8+2	Random	21/50	91.4%
	Spectral	50/50	92.7%

initialization (Table 8). All settings share the same family: sparse layered binary classifiers with symmetric output topology.

6.5 Baselines and Limitations

Logistic regression achieves 100.0% on /a/ vs /i/ and 83.0% on /o/ vs /u/. The Kuramoto network does not match these baselines: a 9-oscillator network with 7 learnable frequencies has far less capacity than a linear classifier on well-separated formant data. The contribution is the gradient identity and the parameter comparison, not classification accuracy.

7 Discussion

7.1 What the Theorem Reveals Beyond EP

The general EP framework guarantees that *some* gradient can be extracted from a contrastive experiment. The specific Kuramoto derivation shows *what* that gradient looks like: it propagates through the network as diffusion on the coupling graph, governed by the Green's function $\tilde{J}^{-1}$ of the graph Laplacian. This connects gradient quality to spectral properties of the coupling topology. The spectral seeding result (Section 6.4) is a first application: using the eigenvectors

of the coupling graph Laplacian to initialize frequencies eliminates the basin failure in all sparse layered settings tested.

7.2 Universality Beyond Sinusoidal Coupling

The proof relies on three structural features: (i) a stable equilibrium, (ii) the implicit function theorem, and (iii) symmetry of the Jacobian. Any coupled oscillator network with an odd coupling function $H(\phi) = -H(-\phi)$ and symmetric coupling strengths $K_{ij} = K_{ji}$ will have a symmetric Jacobian at equilibrium, since $J_{ij} = K_{ij}\, H'(\theta_j^* - \theta_i^*)$ and H' even follows from H odd. Candidate substrates include weakly coupled oscillator models derived from phase response curve theory [Hoppensteadt and Izhikevich, 1999] and biological neural oscillators with arbitrary waveforms, provided the odd-coupling and symmetric-strength assumptions hold. The same IFT argument applies to non-oscillatory equilibrium systems—resistor networks [Dillavou et al., 2024], Hopfield networks [Hopfield, 1982]—wherever the Jacobian is a symmetric graph Laplacian. We have not verified these assumptions for specific physical implementations; doing so is future work.

7.3 Physical Realizability

A physical implementation would map ω_i to bias current or supply voltage, K_{ij} to coupling capacitance, nudging to weak injection current at output oscillators, and gradient readout to phase measurement via time-to-digital conversion or optical interference. In principle, the learning loop would operate at oscillator settling time without a digital backward pass or stored activations. The proof extends to coupled oscillator networks with odd coupling functions and symmetric coupling strengths (Section 7.2), making coupled optical cavities and microring resonators candidate substrates where phase is the native state variable, pending verification of the symmetry and stability assumptions in those specific systems.

7.4 What We Do Not Claim

We do not claim that oscillator networks currently match state-of-the-art digital models on large-scale benchmarks. We do not claim energy efficiency. We do not claim that natural frequency universally outperforms coupling weights—the conditional advantage is specific to sparse architectures and to training-converged seeds; the all-seed comparison is not significant.

We claim: (1) the gradient of a quadratic loss with respect to natural frequencies is physically encoded in the equilibrium phase response, and this encoding is exact under symmetric coupling; (2) conditioned on training convergence, this parameter—explicitly set aside by prior oscillator EP work—outperforms coupling weights on sparse architectures; (3) the $\sim$50% failure rate under random initialization is a property of the loss landscape, not of the gradient computation; spectral seeding eliminates this failure mode in all sparse layered settings tested (100/100 and 50/50 seeds exceeding 60% final test accuracy).

8 Conclusion

A coupled oscillator network at equilibrium already contains its own gradients. Nudge the output, and the phases shift by exactly the amount needed to descend the loss landscape. The Jacobian is a graph Laplacian; its inverse mediates gradient propagation as diffusion. Three independent methods confirm the identity to machine precision across networks of 6 to 200 oscillators. Natural frequency—a parameter explicitly set aside by prior oscillator EP work—is the more effective learning target on sparse architectures. Spectral seeding, which initializes frequencies from the eigenvectors of the coupling Laplacian, eliminates the random-initialization basin failure in all sparse layered settings tested (46% $\rightarrow$ 100% of seeds exceeding 60% final test accuracy on

the primary task; 50/50 on a second task, K-only training, and a larger network). The proof extends to any coupled oscillator network with an odd coupling function and symmetric coupling strengths, suggesting applications beyond Kuramoto dynamics to other physical substrates where phase is the native state variable.

The backward pass was there all along. It was the forward pass, run twice, under symmetric coupling.

Reproducibility

All code, data, and experiment scripts are available in the `phasegrad` package (`https://github.com/caliburlabs/phasegrad`). The verification suite can be run with `pytest tests/`. Experiment result files (JSON) are in `phasegrad/experiments/`. The complete results inventory, including all audit reports, is in `paper/RESULTS_INVENTORY.md`.

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